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Modelling of annular flow through pipes and T-junctions

D. Adechy, R.I. Issa *

Department of Mechanical Engineering, Imperial College, Exhibition Road, London SW7 2BX, UK

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Abstract

This paper describes a computational model for the prediction of isothermal annular two-phase flow in vertical and horizontal intersecting pipes and presents the results of its application to predict phase separation in a horizontal T-junction. The method is based on computational fluid dynamics techniques (CFD) to compute the dispersed core flow simultaneously with the flow of the liquid film along the walls. The core is represented as a dispersed two-phase mixture in which the droplets are tracked using a Lagrangian technique; the wall liquid film is modelled as a thin boundary layer. Full account is made of the interaction between the wall-film and the droplet flow in the core through mass and momentum transfer mechanisms. The method has already been validated against experimental data for fully developed annular flow in vertical and horizontal pipes; there, the predicted film thickness was found to be in satisfactory agreement with data obtained from the literature. The method is now applied to the phenomenon of phase separation occurring in T-junctions where it is found that the predictions for phase split agree quite well with measurements from an independent experiment for a range of phase split ratios. The predictions for wall-film thickness also exhibit similar trends to the data but do not quite match the locations where wall-film thickness peaks occur.

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1. Introduction

Interest in annular flow stems from its common occurrence in many applications, including nuclear reactors and oil and gas production. Although of prime interest are usually the parameters relevant to the transport of fluids in straight pipes (whether horizontal or vertical), these days interest goes beyond that to include intersections in pipeline networks such as T-junctions where it is known that phase segregation occurs. Indeed, the idea of using T-junctions as practical phase

* Corresponding author. Tel.: +44-20-7594-7038; fax: +44-20-7823-8845.
E-mail address: r.issa@ic.ac.uk (R.I. Issa).

Nomenclature

Latin letters

A_d	surface of a droplet (m^2)
C, C_0, C^*	coefficients
C_D	drag coefficient
C_1, C_2, C_μ	turbulence model constants
D	deposition rate ($\text{kg m}^{-2} \text{s}^{-1}$)
D^*	deposition rate parameter ($\text{kg m}^{-2} \text{s}^{-1}$)
d	droplet diameter (m)
dP	pressure gradient term (m)
d_{32}	Sauter mean diameter
$d_{v\mu}$	volume median diameter (m)
E	entrainment rate ($\text{kg m}^{-2} \text{s}^{-1}$)
erfc^{-1}	inverse complementary error function
F	coefficient
f	friction factor
$\vec{F}_{DR}$	total drag force (N)
$\vec{F}_{GR}$	body force due to the gravity (N)
$g, \vec{g}$	gravity and gravity vector (m s^{-2})
k	turbulent kinetic energy ($\text{m}^2 \text{s}^{-2}$)
k_s	equivalent sand roughness (m)
$\dot{M}_{LE}$	mass flux of entrained droplets ($\text{kg m}^{-2} \text{s}^{-1}$)
m_d	mass of a droplet (kg)
N_r	normally distributed random number
p	pressure (Pa)
R	pipe radius (m)
Re_c	core Reynolds number
Re_d	droplet Reynolds number $Re_d = \rho_G d \bar{u}_G - \bar{u}_d / \mu_G$
S_{u_G}	momentum source term for the gas ($\text{kg m}^{-2} \text{s}^{-2}$)
S_m	source term for the δ transport equation ($\text{kg m}^{-2} \text{s}^{-1}$)
t	time (s)
$u, \vec{u}$	velocity and velocity vector (m s^{-1})
U	superficial velocity (m s^{-1})
u_G^*	gas phase friction velocity $u_G^* = \sqrt{\tau_{IF} / \rho_G}$ (m s^{-1})
$\bar{u}_G$	mean value of gas phase velocity (m s^{-1})
$\bar{u}$	averaged liquid film velocity $\bar{u} = \frac{1}{\delta} \int_0^\delta u dX_3$ (m s^{-1})
$\bar{u}_d$	mean value of droplet velocity (m s^{-1})
We_ϕ	Weber number based on the tube diameter $We_\phi = \rho_G U_G^2 \phi / \sigma$
X	co-ordinate (m)

Greek letters

μ	dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
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ν	kinematic viscosity ($\text{m}^2 \text{s}^{-1}$)
ρ	density (kg m^{-3})
φ	volume fraction of the liquid film
β	coefficient
ϕ	pipe diameter (m)
δ	liquid film thickness (m)
δ^*	displacement thickness (m)
θ	angle from the vertical
ε	rate of dissipation ($\text{m}^2 \text{s}^{-3}$)
κ	Von Karman constant for single-phase flows
χ	variable
σ	surface tension (N m^{-1})
$\sigma_k, \sigma_\varepsilon$	turbulent Schmidt numbers
τ	shear stress (N m^{-2})

Subscripts

All unsubscripted quantities refer to the liquid film

G, C, d gas phase core and droplet respectively

1, 2, 3 axial, circumferential and transverse directions

Superscripts

1, 2, 3 axial, circumferential and transverse directions

' random perturbation

eff effective

if gas–liquid interface

l, t laminar and turbulent

T transposed

w wall

separators has been mooted for some time. For such concept to succeed, not only is it essential to develop an understanding of how the degree of phase separation depends on the flow conditions but it is also important to be able to model the flow in such configurations in order to predict phase segregation. This paper proposes a model to compute the relevant parameters in annular flow in straight pipes in general and to predict phase separation in T-junctions in particular.

Annular flow, in both horizontal and vertical pipes has been the subject of many theoretical studies. However, most of these studies have focused on fully developed flow, since the phenomena involved in this flow regime are quite complex. First there are waves on the interface of the liquid-film; second there is the phenomenon of droplet entrainment from the film into the core, and their deposition back onto the film. Then there is the wetting of the pipe top in horizontal flow by the liquid film against the action of gravity, which should in principle drain the film to the pipe bottom.

To predict the film thickness in annular flows under fully developed flow conditions, many authors (see [1–5]) focused on the liquid film transport equations (i.e. mass and momentum). The

effect of the gas on the film was accounted for via empirical correlations or by using actual measured values. Since in those approaches the transport equations for the gas phase are not solved, only weak one-way coupling between the gas flow and the liquid film is accounted for (via the modification of interfacial shear stresses for the film thickness calculation). More recently Tso and Sugawara [6] used a two-dimensional Eulerian approach (three-fluid model) to effect simultaneous calculation of the liquid film, the droplet and the gas flow fields. Although their model enabled coupled calculations between the phases, the film interface could not be accurately located (due to numerical smearing). In contrast, the present approach is three-dimensional and assumes a sharp interface between the liquid film and the core.

Over the years, various mechanisms have been proposed to explain the presence of a liquid film at the top of tubes in horizontal annular flows. These comprise: the secondary flow mechanism, the transport of liquid by entrainment and deposition of droplets, the spreading by surface tension forces and the spreading by waves. Authors like Lin et al. [2], Laurinat et al. [3] and Jackman et al. [7] identified the secondary flow mechanism as the major factor for lifting the liquid against gravity. Hurlburt and Newell [40] extended the model of Laurinat et al. and developed a balance between the normal Reynolds stresses in the circumferential direction within the film and the component of the gravitational force in that direction. That balance was used to predict the film thickness distribution in horizontal pipes. Fukano and Ousaka [5] suggested that the phenomenon is due to a pressure gradient opposing gravity inside waves of large amplitudes (to which they referred as the ‘pumping action of disturbance waves’). Recently, Fukano [8] used an interface tracking technique to simulate the formation of waves in annular flow and demonstrated the wave action that was proposed in his earlier work.

In the present work, the wave inclination mechanism suggested by Butterworth [9] and recently verified by the experiments of Jayanti et al. [10] and Paras and Karabelas [11] is adopted to explain the presence of liquid at the top of pipes. In this theory, the form drag associated with the gas impacting on inclined waves, leads to a lift force opposite to gravity. A model to represent this force is proposed herein and is incorporated into the model.

Over the years several attempts have been made to predict phase separation at T-junctions using different approaches: empirical, phenomenological, and computational fluid dynamics (CFD—two-fluid models). In the empirical approach of Seeger et al. [13] measurements of phase separation at T-junctions were conducted in air–water and steam–water systems; from these direct empirical correlations for different branch orientations were derived. The aim was to produce engineering purpose correlations, which could be used in the design of pipe junctions. Inevitably, such correlations are not suitable for general application (since they are flow-pattern and geometry dependent). Chien [14] also used an empirical approach to determine the phase separation, deriving a phase-splitting equation. In that case the model was applicable to any flow condition.

Phenomenological approaches on the other hand are more general since they are based on theoretical grounds. Based on experimental observations and theoretical considerations Roberts et al. [12], Azzopardi and Whalley [15], Shoham et al. [16], Hwang et al. [17], Hart et al. [18], Lemmonier and Hervieu [19], McCreery and Banerjee [20], Lightstone et al. [21], Lahey [22], Roberts et al. [23], Peng and Shoukri [24] developed phenomenological models to predict the flow split at junctions. Azzopardi and Whalley [15] suggested that the gas entering the branch of the junction drags the liquid film with it, and introduced an apparent angle over which the liquid is extracted. Based on experimental data they could find a linear correlation between this angle and

the air flow rate entering into the branch. Shoham et al. [16] presented two models, one was purely geometrical: for a given a gas-split ratio, a uniform film thickness around the pipe circumference was calculated and the liquid split was obtained from geometrical consideration. The other, more elaborate model assumed that the preferred liquid route was dictated by the net centripetal force acting on the liquid. With knowledge of the gas fraction flowing into the side arm, the streamline of the liquid phase directed into the side arm could be determined. The authors introduced the idea of two distinct dividing streamlines, one for the gas phase (input) and one for the liquid (calculated). Such an idea about dividing streamlines was used later by others such as: Hwang et al. [17], Hart et al. [18], Lahey [22], Peng and Shoukri [24]. Hwang et al. [17] determined the streamlines of the liquid by solving the Euler equations in the streamwise and normal directions and could thus deduce the fraction of liquid taken off in the branch. Lahey [22] and Peng and Shoukri [24] used a similar approach. Hart et al. [18] also introduced the idea of dividing streamlines and in their double stream model used a one-dimensional approach and solved the Bernoulli equations for the two-phases. Lemmonier and Hervieu [19] also used a methodology based on dividing streamline calculations. From potential flow theory they first predicted the flow of the continuous phase and thereafter the streamlines of the dispersed phase; their approach accounted only for one-way coupling.

Most of the CFD work on phase separation in T-junctions has been carried out for either bubbly or particulate (i.e. particles in gas) flows. Thus, Lahey [22,25], Issa and Oliveira [26], Ellison et al. [27] used the two-fluid model where both gas and liquid phases were considered as fully coupled continuous media. They solved the multi-dimensional transport equations for the two phases numerically, and accounted for turbulence transport through turbulence models. Hatziaivramidis et al. [28] also used the two-fluid model in which the flow pattern resulted from the numerical solution itself and therefore the annular flow regime could in principle be accounted for, although it could not account for the mass transfer between the film and the core due to entrainment and deposition.

The present approach is based on the solution of transport equations for the dispersed droplet and gas phases in the core, together with simplified equations for the liquid film, using finite volume CFD techniques. A Lagrangian approach is employed to predict the trajectories of droplets through the gas core, which is assumed to behave as a continuum. The methodology accounts fully for the interaction between the phases including mass exchange due to entrainment and deposition.

2. Methodology

The basic methodology employed here comprises two elements. In the first, the core flow is modelled as a continuous phase (the gas) carrying a dispersed liquid phase in the form of droplets. This core flow is computed by solving the three-dimensional transport equations (i.e. the Reynolds averaged Navier–Stokes equations) for the gas phase, while the droplets are tracked by solving their equations of motion in a Lagrangian way. The second element in the methodology is a model for the liquid film flowing along the wall. Here the film is assumed to be thin and the flow within it is of the boundary layer type. The conservation equations for momentum and mass are thereby much simplified and are solved to capture the evolution of the wall-film. To account for

the interaction between the core and wall-film flows, the two models are coupled together whereby mass and momentum transfer taking place between the two regions, are linked to local flow conditions. The wavy liquid film interface is assumed to act as a stationary rough surface (the liquid film velocity is in general much lower than the gas velocity) in relation to the core flow. In order to model the effect of this rough surface on the dispersed flow, a roughness parameter is applied locally to each computational cell adjacent to the wall. To simulate the entrainment process, droplets are assumed to originate from the wall-film and are injected in the solution domain using available correlations (for initial ejection velocities and entrainment rates). Droplet deposition on the film is also accounted for in the model. The interface therefore acts as a boundary condition in terms of momentum and mass transfer.

It should be emphasised that the assumption of very thin liquid film is central to the method (the flow area of the core is taken to be constant and remains unaffected by the film regardless of its thickness). This assumption naturally limits the applicability of the technique to situations where the film thickness is not greater than a few percent of the pipe.

2.1. The liquid film model

A wall-film model similar to that of Ahmadi-Befrui et al. [29], which was originally developed for spray impingement on walls is developed here for application to annular flow. In the model, the liquid film is assumed to be thin, two-dimensional (axial and circumferential directions) and to behave like a boundary layer, so that the dominant gradients are in the direction normal to the flow. The shear stress across the film is assumed to be constant and surface tension effects are neglected. Inertial effects are accounted for approximately. A one-seventh-power law velocity profile is assumed to prevail across the film whose surface is also assumed to be locally parallel to the solid wall surface.

The transport equations for mass and momentum are first written in the general co-ordinate shown in Fig. 1, which maps the wall region. Those equations are then integrated across the film in the direction normal to the wall (using the assumed velocity profile and taking the velocity normal to the wall to be zero) to reduce them to a two-dimensional form, the two dimensions being in the axial and circumferential directions (Fig. 2). Details of the basic equations and the integration process are given by Adechy [30] and are therefore omitted here.

The continuity equation resulting from integration across the film, reduces to a transport equation for the film thickness ' δ ' as follows:

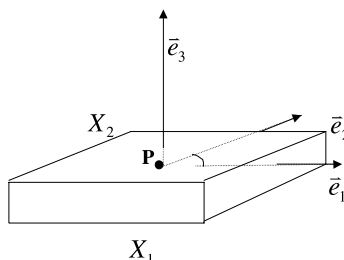


Fig. 1. Co-ordinate system for the liquid film.

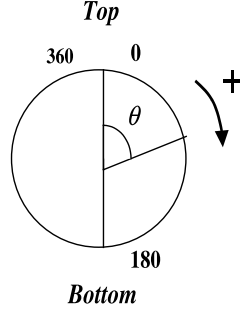


Fig. 2. Circumferential position.

$$\frac{\partial}{\partial t}(\rho\delta) + \frac{\partial}{\partial X_1}(\rho\bar{u}_1\delta) + \frac{\partial}{\partial X_2}(\rho\bar{u}_2\delta) = D - E \quad (1)$$

where $\bar{u}_1$ and $\bar{u}_2$ are the average velocity components in the axial and tangential directions respectively, D is the rate of deposition of droplets on the film and E is the rate of entrainment of droplets from the film.

The rate of entrainment of droplets from the film into the core, E , needs to be specified. There are many correlations available in the literature to model this effect and naturally the results of the simulations presented herein would, to some extent, depend on the model chosen. However, the simulations have shown that the results are more influenced by the interfacial force model described below than by the entrainment process. Here, we chose to use the entrainment rate proposed by Ueda [31] that was determined from experimental data and based on the force balance at the interface between the liquid film and the gas. The reason behind choosing this correlation is the physical basis used in its derivation, together with the fact that it gave the right entrainment rates in earlier validation studies in horizontal pipes (see [36]). The correlation is given by:

$$\begin{cases} E = 3.54 \times 10^{-3} \chi^{0.57}, & \text{if } \chi > 120 \\ E = 0, & \text{otherwise} \end{cases} \quad (2)$$

The variable χ is defined by:

$$\chi = \frac{|\tau_1^{\text{if}}|}{\sigma} \left(\frac{\varphi \bar{u}_1}{\sigma} \right)^{0.6} \quad (3)$$

where τ_1^{if} stands for the interfacial shear stress (gas–liquid) in the main direction of the flow and the liquid film hold-up φ is expressed as:

$$\varphi = 1 - \left(1 - \frac{\delta}{R} \right)^2 \quad (4)$$

It should be pointed out that the Ueda correlation for E relates to developed flow rather than a developing one as is the case in T-junction flows. Its use here is justified on the grounds that the departure of the entrainment rate from its equilibrium level is a second order effect compared to

other major phenomena such as the circumferential force that maintains the liquid film presence on the pipe wall. Moreover, to use an alternative correlation pertaining to a developing flow could make the present overall model specific to the particular flow for which the correlation was derived.

The deposition rate D is determined from the Lagrangian model for the core flow (to be described shortly): those droplets which reach the wall-film (as determined by the Lagrangian droplet tracking scheme used) are assumed to be perfectly absorbed.

The momentum equations for the liquid in the film are expressed as:

$$\frac{\partial}{\partial t}(\rho u_1) + \frac{\partial}{\partial X_1}(\rho u_1 u_1) + \frac{\partial}{\partial X_2}(\rho u_2 u_1) - \frac{\partial}{\partial X_3}(\tau_1) = -\frac{\partial p}{\partial X_1} + \rho g_1 \quad (5)$$

in the axial direction and,

$$\frac{\partial}{\partial t}(\rho u_2) + \frac{\partial}{\partial X_1}(\rho u_2 u_1) + \frac{\partial}{\partial X_2}(\rho u_2 u_2) - \frac{\partial}{\partial X_3}(\tau_2) = -\frac{\partial p}{\partial X_2} + \rho g_2 \quad (6)$$

in the circumferential direction.

The velocity components u_1 and u_2 appearing in the above equations, are now assumed to vary in the X_3 direction (normal to the film) according to the one-seventh power profile. Thus

$$\begin{cases} \frac{u_1}{\bar{u}_1} = \frac{8}{7} \left(\frac{X_3}{\delta} \right)^{1/7} \\ \frac{u_2}{\bar{u}_2} = \frac{8}{7} \left(\frac{X_3}{\delta} \right)^{1/7} \end{cases} \quad (7)$$

where $\bar{u}_1$ and $\bar{u}_2$ are the respective average velocities in the film. Eqs. (5) and (6) are then cast in terms of $\bar{u}_1$ and $\bar{u}_2$ by integration across the film before being discretised using the finite volume method (details are given by Adechy [30]). Solution of the resulting equations yields the distributions of $\bar{u}_1$ and $\bar{u}_2$ over the film. In solving the above equations, values of the pressure gradients are needed: they are obtained from the solution of the core flow, which is assumed to determine the pressure field.

The stresses τ_1 and τ_2 in Eqs. (5) and (6) may be defined as:

$$\tau_1 = \mu^{\text{eff}} \frac{\partial u_1}{\partial X_3} \quad (8)$$

and

$$\tau_2 = \mu^{\text{eff}} \frac{\partial u_2}{\partial X_3} \quad (9)$$

where μ^{eff} is an effective viscosity to be defined later. The values of τ_1 and τ_2 can thus be obtained by differentiating the velocity profiles in expressions (7).

In the transverse direction, the momentum equation reduces to:

$$-\frac{\partial p}{\partial X_3} + \rho g_3 = 0 \quad (10)$$

In the above equations, the effect of droplet splashing onto the liquid film is neglected.

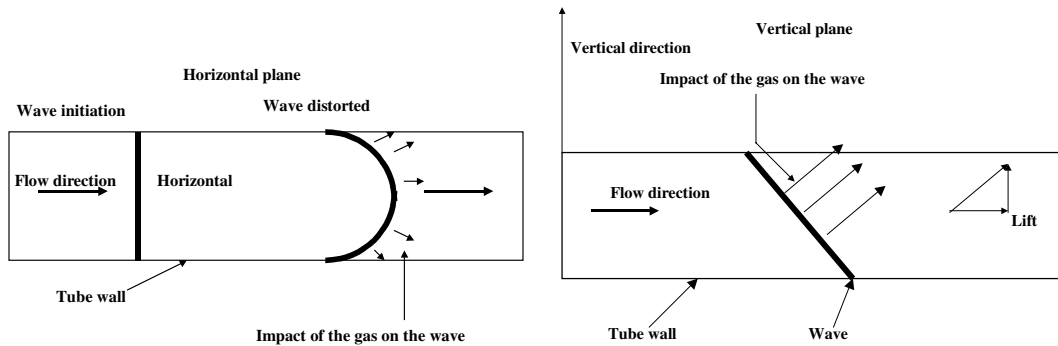


Fig. 3. Wave action mechanism horizontal and vertical plane.

The model for the interfacial momentum transfer between the gas phase and the liquid film in the circumferential and axial directions is now described. In horizontal annular flow, it is well known that a liquid film, albeit a thin one, is usually present at the top of the pipe. Thus, there must be a force that maintains this liquid against the action of gravity, which would otherwise tend to drain the liquid to the bottom. According to the hypothesis of Butterworth [9], waves propagating on the surface of the film in horizontal flow are inclined with respect to the axial direction (see Fig. 3). Since the gas travels much faster than the liquid, it impacts on the waves thereby generating form drag. Since the wave is inclined to the direction of the flow, this form drag will have two components: one in the axial direction, and the other in the circumferential direction (opposing gravity). In the present approach, this form-drag force is represented as a steady force (although in reality it is time varying corresponding to surges of surface waves) that augments the circumferential shear stress due to interfacial friction. Thus an additional stress τ_2^{if} regarded as originating form drag exerted by the gas impacting on inclined waves is now introduced.

Naturally, the wave induced form-drag force proposed by Butterworth [9] must be a function of several quantities like the characteristic wave height and the relative velocities between the film and core. A proper correlation for this force is therefore needed. However, in the absence of such correlation, which has to be derived from experimental data, an ad-hoc relationship has to be used here. According to the experimental results (see Fig. 7), there is a small drainage of liquid from the top to the bottom of the pipe; therefore τ_2^{if} must be slightly smaller than the gravity term. Moreover in the wave inclination mechanism, the force exerted by the gas on the waves is a contact force and the inclination of the waves (deviation from the vertical) depends on the strength of this contact force (see the experimental results of Paras and Karabelas [11]). That suggests that for coherent waves to exist, the vertical component of the impacting gas force must be directly proportional to the gravity force (side view in Fig. 3). In order to find a simple expression for this additional circumferential shear τ_2^{if} , the following force-balance in the circumferential direction is drawn up:

$$\tau_2^{\text{if}} - \tau_2^{\text{w}} = -\rho g_2 \delta \quad (11)$$

In Eq. (11) the pressure gradient term which is of $o(\delta^2)$ compared to the gravity term which is $o(\delta)$, is neglected.

Now the wall shear stress is given by

$$\tau_2^w = \tau_2^\delta = \mu^{\text{eff}} \left(\frac{\partial u_2}{\partial X_3} \right)_{X_3=\delta} \quad (12)$$

and with the assumption of one-seventh-power law for the velocity profile, the form of τ_2^{if} can be found. Thus

$$\tau_2^{\text{if}} \cong \frac{u_2^{\text{if}} \mu^{\text{eff}}}{7\delta} - \rho g_2 \delta \quad (13)$$

where u^{if} is the value of the u_2 velocity component at the film surface and is uniquely related to the average velocity $\bar{u}_2$ by the assumed velocity profile. The final form of the expression is obtained by replacing g_2 (the component of gravity in the circumferential direction) by $g \sin \theta$:

$$\tau_2^{\text{if}} \cong \frac{u_2^{\text{if}} \mu^{\text{eff}}}{7\delta} - \rho g \sin \theta \delta \quad (14)$$

in which the first term can be neglected.

Concerning the axial momentum transfer, an interfacial friction factor for a completely rough regime together with the formulation of Wolf [32] for the sand roughness are utilised to determine the interfacial stress τ_1^{if} . As previously indicated, the intermittence of the waves is only taken into account in an average sense. Thus

$$f_1^{\text{if}} = \frac{1}{4} \frac{1}{\left(2 \log \frac{R}{k_s} + 1.74 \right)^2} \quad (15)$$

where

$$k_s = 7.0623\delta - 0.0088\phi \quad (16)$$

and ϕ is the pipe diameter.

The effective viscosity in Eq. (13) is based on the fully turbulent viscosity (derived from the Von Karman velocity profile) and is expressed as

$$\mu_t = \mu \left(\gamma \frac{\delta^+}{2.5} - 1 \right), \quad \delta^+ \geq 2.5, \quad \mu_t = 0 \quad (17)$$

where

$$\delta^+ = \sqrt{\frac{\tau_1^{\text{if}}}{\rho}} \frac{\delta}{\nu} \quad (18)$$

The coefficient γ in Eq. (17), and whose value is chosen to be 1.3, is an additional parameter that represents the influence of waves at the interface of the liquid film. This approach has been proposed by Jackman et al. [7] who used an apparent wave viscosity in their annular flow model to account for the enhancement of momentum transfer near the interface due to the waves. The main assumption here is that the enhancement of turbulent viscosity by the waves is similar to that due to the turbulent eddy motion.

Finally, the effective viscosity is expressed as

$$\mu^{\text{eff}} = \mu + \mu^t \quad (19)$$

and is used in the calculation of the stresses from expressions (8) and (9).

2.2. The core model

The core flow is modelled as a continuum phase (gas) carrying a number of discrete droplets. Eulerian transport equations are therefore solved for the gas phase; turbulence is modelled by the standard k – ε model. The dispersed droplets are followed along their trajectories in a Lagrangian approach. This core model already exists and is incorporated in the commercial software STAR-CD which was used for the purpose.

The transport equations of the gas phase are expressed in their continuum form as follows:

Continuity

$$\frac{\partial}{\partial t} \varphi_G \rho_G + \nabla \cdot \rho_G \varphi_G \bar{\mathbf{u}}_G = 0 \quad (20)$$

Momentum

$$\frac{\partial}{\partial t} \varphi_G \rho_G \bar{\mathbf{u}}_G + \nabla \cdot \rho_G \varphi_G \overline{\mathbf{u}_G \mathbf{u}_G} = -\varphi_G \nabla p + \varphi_G \rho_G \mathbf{g} + \nabla \cdot \varphi_G (\boldsymbol{\tau}_G^l + \boldsymbol{\tau}_G^t) + \varphi_G S_{u_G} \quad (21)$$

where ρ_G , φ_G , $\bar{\mathbf{u}}$ are the mean values of density, volume fraction and velocity respectively. The source term S_{u_G} accounts for the effects of the droplets on the gas phase. Also, through this source term, the additional form drag force at the boundary between the core and the film (due to the previously discussed impact of the gas on inclined waves) is introduced.

The k and ε equations pertaining to the turbulence model are respectively:

$$\frac{\partial}{\partial t} \varphi_G \rho_G k + \nabla \cdot \rho_G \varphi_G \bar{\mathbf{u}}_G k = \nabla \cdot \left(\frac{\varphi_G \mu_G^{\text{eff}}}{\sigma_k} \nabla k \right) + B - \rho_G \varphi_G \varepsilon \quad (22)$$

and

$$\frac{\partial}{\partial t} \varphi_G \rho_G \varepsilon + \nabla \cdot \rho_G \varphi_G \bar{\mathbf{u}}_G \varepsilon = \nabla \cdot \left(\frac{\varphi_G \mu_G^{\text{eff}}}{\sigma_\varepsilon} \nabla \varepsilon \right) + \frac{\varepsilon}{k} (C_1 G - C_2 \varphi_G \rho_G \varepsilon) \quad (23)$$

with

$$B = \varphi_G \mu_G^t (\nabla \bar{\mathbf{u}} + \nabla \bar{\mathbf{u}}^T), \quad (24)$$

$$\mu_G^t = \frac{C_\mu \rho_G k^2}{\varepsilon} \quad (25)$$

and

$$\mu_G^{\text{eff}} = \mu_G^l + \mu_G^t \quad (26)$$

It should be pointed out that the turbulence model used pertains to single phase flow. This implies that modification of the turbulence field due to the presence of droplets is neglected. This

may be justifiable for dilute droplet flow, but would not be appropriate if the droplet phase fraction became high. In such a case, proper extension of the $k-\varepsilon$ model to account for two-phase effects must be made.

For the dispersed phase, a Lagrangian treatment is chosen. With the aid of a vector notation for the absolute velocities and using material derivatives in time, the continuity and momentum equations for each particle are expressed respectively as:

$$\frac{dm_d}{dt} = 0 \quad (27)$$

and

$$m_d \frac{d\vec{u}_d}{dt} = \vec{F}_{DR} + \vec{F}_{GR} \quad (28)$$

The two forces on the right-hand side of Eq. (28) represent the two most important effects, which are the drag and gravitational forces. The drag is taken as

$$\vec{F}_{DR} = \frac{1}{2} C_D \rho A_d |\vec{u}_G - \vec{u}_d| (\vec{u}_G - \vec{u}_d) \quad (29)$$

where $\vec{u}_G$ and $\vec{u}_d$ are the velocity vectors of the gas and droplet respectively. The drag coefficient C_D is expressed as follows:

$$C_D = \begin{cases} \frac{24}{Re_d} (1 + 0.15 Re_d^{0.687}) & Re_d \leq 10^3 \\ 0.44 & Re_d > 10^3 \end{cases} \quad (30)$$

The gravity force is

$$\vec{F}_{GR} = m_d \vec{g} \quad (31)$$

For droplet transport, turbulence is accounted for by using a stochastic approach (as incorporated in STAR-CD). It is assumed that each droplet interacts with the gas eddies in a random way and the perturbation velocities u'_G are calculated as follows:

$$u'_G = \left(\frac{2k}{3} \right)^{1/2} \text{sign}(N_r) \text{erfc}^{-1}(N_r), \quad N_r \in [-1, 1] \quad (32)$$

Droplet entrainment takes place continuously from the film surface; hence the newly entrained droplets must be launched into the core computation. The initial ejection velocity of the droplets u_{3d} is calculated from the formulation of Andreussi and Azzopardi [33] as follows:

$$u_{3d} = 12 \sqrt{\frac{\rho_G}{\rho}} u_G^* \quad (33)$$

The initial velocities of the droplets in the axial and circumferential directions are assumed to be those of the liquid film at the interface (i.e. u_1^{if} , u_2^{if}).

The droplet diameter features in the above equations and, obviously, the entrained droplets will have a range of values for this diameter. However, within the context of the present model, which relies on several empirical correlations, it is felt that to account for the variability of the droplet size is not warranted. This however is not an inherent limitation of the methodology; the size distribution can easily be taken into account if desired. Here, a single value for the volume median

diameter defined by Tatterson et al. [34] is used as a characteristic droplet diameter. In the cited work, the volume median diameter was correlated to the Sauter mean diameter by

$$\frac{d_{32}}{d_{v\mu}} = 0.7 \quad (34)$$

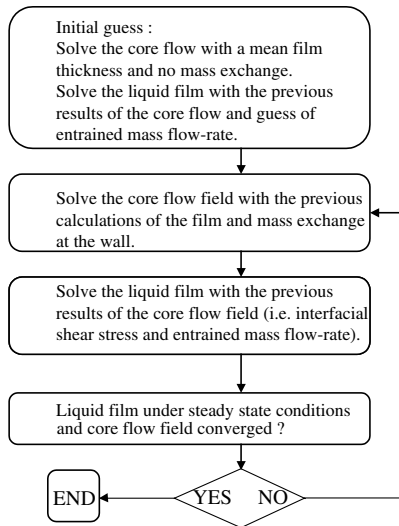
The equation given by Ambrosini et al. [35] provides the value of the Sauter mean diameter as

$$\frac{d_{32}}{\delta} = 2.2 \left(\frac{\sigma}{\rho_G f_1^{\text{if}} U_G^2 \delta} \right)^{0.5} \left(\frac{\rho_G}{\rho} \right)^{0.83} \exp \left(\frac{0.6 \dot{M}_{\text{LE}} \phi}{\rho U_G d_{32}} + \frac{99}{W_{e\phi}} \right) \quad (35)$$

In the present paper, mean values are used to compute the initial ejection velocities of droplets and droplet diameter (i.e. each ejected droplets will have the same ejection velocity and the same mean diameter). In the formulation of d_{32} and $d_{v\mu}$, the interfacial friction terms are calculated according to the pressure drop in the pipe.

2.3. Method of solution

The liquid film equations are solved by marching explicitly in time until the steady state is reached, while the core is solved for implicitly assuming a steady state. The liquid film and the core phases are computed in a predictor–corrector manner as shown in the diagram given below.



3. Results and discussion

3.1. Horizontal annular flow

The method was first applied to the case of fully developed horizontal pipe flow for validation purposes and the results of that study have been presented by Adechy and Issa [36]; some of the

results are re-presented here. In that work, the predictions were compared with the well-known experimental data of Butterworth and Pulling [37] for a 32 mm internal diameter pipe. The inlet conditions are 0.051 kg s^{-1} for air and 0.064 kg s^{-1} for the water flow-rates. The test section is at a pressure of 2.5 bar. The computational mesh is composed of 89677 cells (161 cells in the cross-section and 557 rows in the axial direction). This mesh is deemed to be sufficient to give reasonable numerical accuracy since the flow does not feature large gradients in the axial direction. Concerning the entrainment process, as the number of dispersed elements is large, a sample of the total population is analysed. Each member of the sample is a ‘computational droplet’ representing a group of real elements. In the present test case, 2875 ‘computational droplets’ each containing 400 to about 700 real droplets were utilised.

Fig. 4 shows experimental and calculated circumferential values of the film thickness from top to bottom of the pipe. The theoretical values are those of the present study and of the model of Tso and Sugawara [6]. The experimental and calculated values of the film thickness are in good agreement but somewhat better results are obtained with the present model than Tso and Sugawara’s model in the lower part of the pipe.

Fig. 5 shows the evolution of the film thickness in the pipe with axial distance for different circumferential locations around the pipe. A fully developed flow in the liquid is asymptotically reached after about 60 pipe diameters, which is quite below the usually quoted experimental value

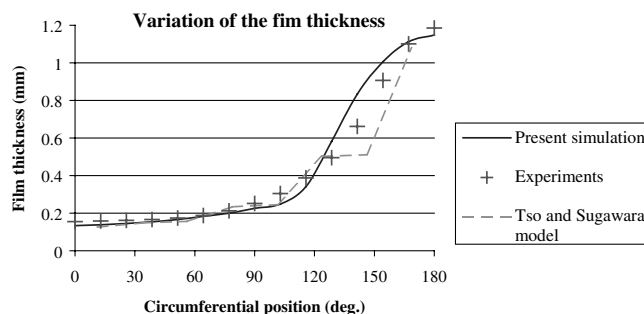


Fig. 4. Circumferential variation of the film thickness, horizontal annular flow.

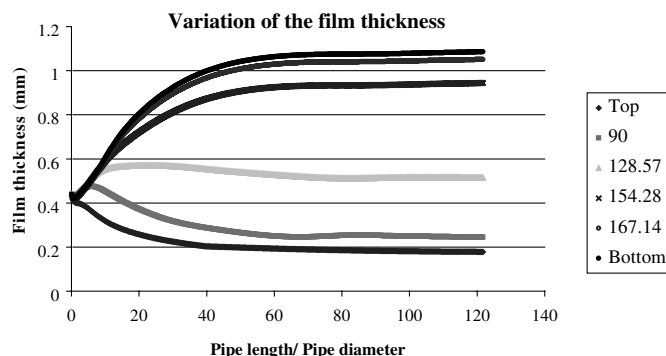


Fig. 5. Axial variation of the film thickness.

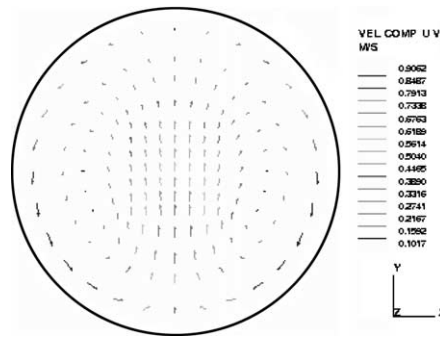


Fig. 6. Circumferential motion of the gas.

of around 140 pipe diameters. This discrepancy could be explained by the fact that the local correlations used in the liquid film model are those established for fully developed flow conditions but are applied here everywhere.

Fig. 6 shows the circumferential motion of the gas phase, which results from the imposition of the lift force. It is found that the maximum circumferential gas velocity is about 3.5% of the maximum axial gas velocity, which shows that this secondary motion is quite weak. It should be noted that the secondary gas motion is in a direction opposite to that of the secondary flow theory, where the circumferential gas motion is supposed to transport the liquid to the top of the pipe. Indeed, in their experimental study of stratified flow in a horizontal pipeline, Dykhno et al. [38] showed that in the ‘stratified-annular’ regime gas flow directed downward at the wall and upward at the centre of the pipe can occur, as found in the present study. It should be remarked that if the secondary gas motion was itself responsible for lifting the liquid film to the pipe top, then it is estimated that secondary velocities of the order of 9 m s^{-1} are needed to provide that force. Such large circumferential velocities have never been observed experimentally (see [4,10]) so clearly this mechanism cannot be seriously considered as being responsible for the phenomenon.

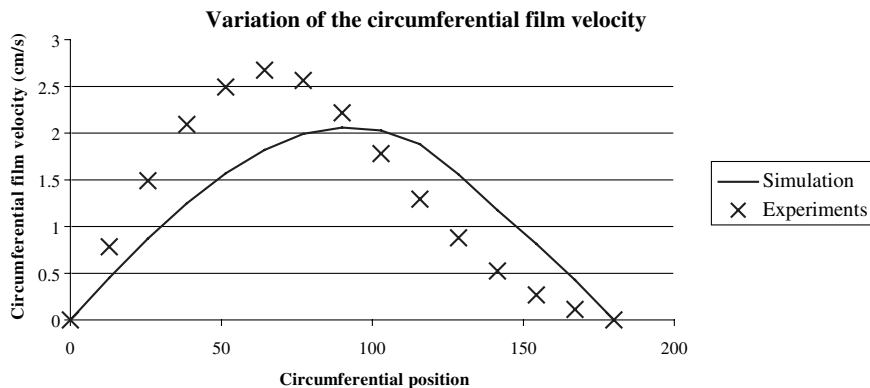


Fig. 7. Variation of the circumferential film velocity.

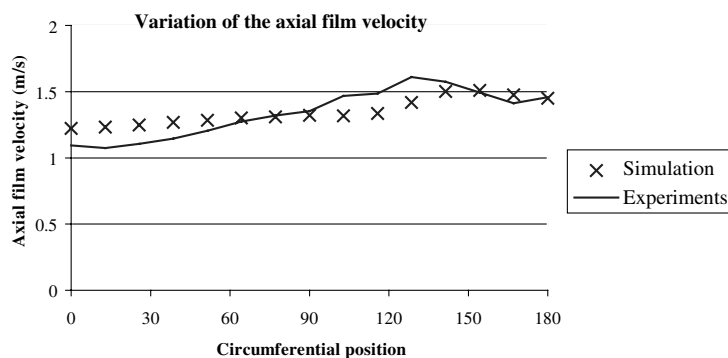


Fig. 8. Circumferential variation of the axial film velocity.

Figs. 7 and 8 give the experimental and calculated values of the averaged circumferential and axial film velocities around the pipe. In those figures, the velocities have been determined from the experimental values of the flow-rates and of the film thickness. Good agreement is obtained between theory and experiment.

The above validation exercise vindicates the use of the Ueda entrainment model, in that when the flow becomes fully developed, the rate of entrainment (given by the Ueda correlation) is balanced exactly by the rate of deposition as determined from the Lagrangian tracking of the droplets, which is of course an independent computation. This indirectly demonstrates the validity of the Ueda model, at least when the flow becomes fully developed.

Finally, the present model predicts the pressure drop in the pipe to within 14% of the measured value.

3.2. Horizontal T-junction flow

Here, the results of predictions for liquid film thickness and liquid split for the flow in a horizontal T-junction are presented and compared with measurements made by Conte and Azopardi [39]. The diameter of 127 mm is the same for the inlet, run and branch pipes of the junction. The inlet flow rates are 0.392 kg s^{-1} for the air and 3.42 kg s^{-1} for the water; the inlet pressure is 1 bar. At the outlets, the split of the gas phase is prescribed. The reason of specifying this ratio rather than the pressure at the outlets (which would be the natural boundary conditions) is that the experimental data actually specified the split ratio of the gas phase.

Two meshes were used in these computations, one with 90,000 cells and the other with about 360,000 cells (Figs. 9 and 10). This was to establish mesh-independence of the calculations. The total number of computational droplets was typically from 3000 to 5000. Those computational droplets represented up to about 4000 real droplets.

The first comparison between the model and experiment concerns the film thickness and is made at one pipe diameter away from the geometrical centre of the junction as shown in Fig. 11 and at different split ratios of the gas phase of 0.3, 0.5 and 0.7 (this always refers to the ratio of the flow rate at the branch to the flow rate at inlet). Figs. 12 and 13 present the convention adopted to locate the circumferential position around the pipe circumference at the inlet, run and branch of the junction.

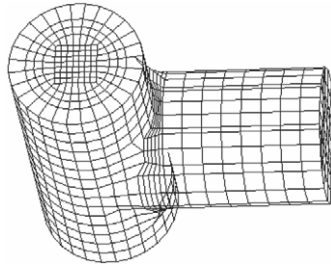


Fig. 9. Coarse mesh.

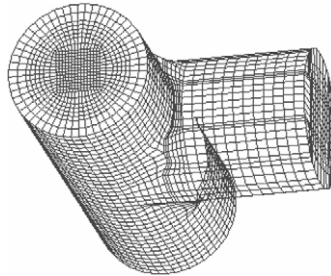


Fig. 10. Fine mesh.

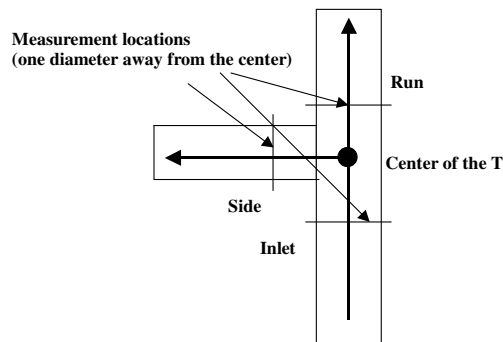


Fig. 11. Film thickness measurement locations at the T.

Figs. 14–16 show the variation of the film thickness around the pipe circumference at the inlet, one pipe diameter away from the centre of the junction for the three different gas split ratios. For those different cases, the model can reasonably reproduce the variation of the film thickness around the pipe circumference. However, it is seen that the calculated thickness is consistently larger than the measured one. Since the flow rate of the liquid should be the same in both calculation and experiment, this suggests that the velocity of the liquid in the film is under-predicted, thereby leading to a bigger film thickness. Here, the one-seventh-power law velocity profile might be failing to reproduce the dynamics of the flow especially at the bottom of the pipe where the film is very thick. Also, the effect of too high a turbulent viscosity in the liquid film model would lead

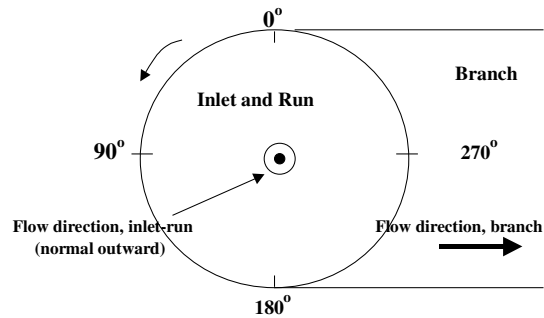


Fig. 12. Convention for the circumferential position: inlet and run.

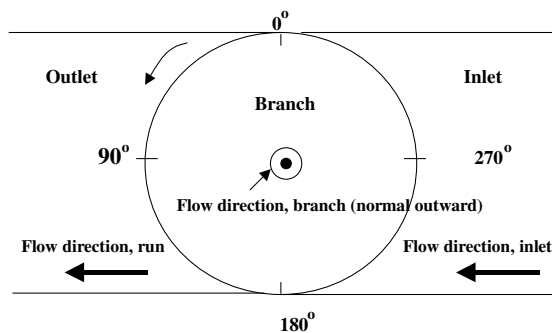


Fig. 13. Convention for the circumferential position: branch.

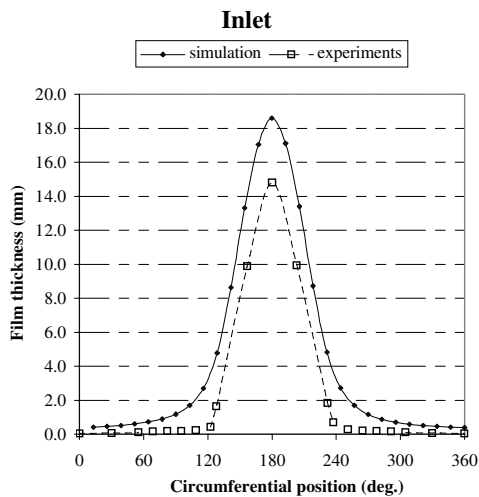


Fig. 14. Circumferential film thickness comparison, inlet, gas split ratio 0.3.

to slowing down of the liquid velocity and therefore to thickening of the film. Alternatively, the rate of entrainment of liquid droplets from the film may be under-predicted, although this could

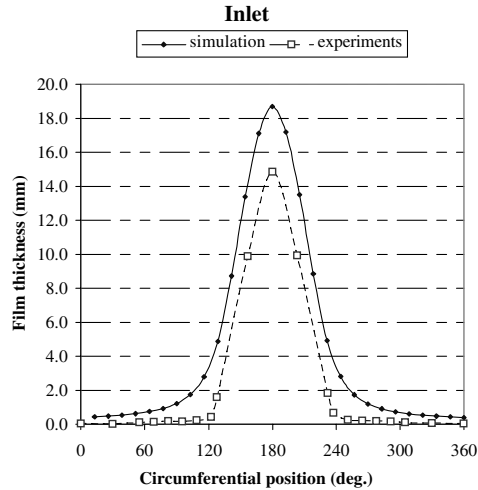


Fig. 15. Circumferential film thickness comparison, inlet, gas split ratio 0.5.

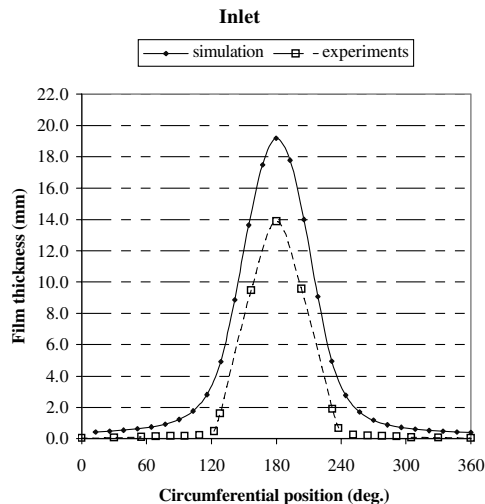


Fig. 16. Circumferential film thickness in inlet at gas split ratio 0.7.

not on its own account for the discrepancy. In any event, in the absence of measurement, it is difficult to quantify the effect of this parameter on the observed discrepancies.

In Figs. 17–19, it can be seen that unlike the experimental results, the calculated film thickness slightly increases as the gas split ratio increases (the location is at the bottom of the pipe). With a big enough gas-split ratio, the liquid film grows markedly forming a blockage in the run, an event for which the assumption of thin film (on which the model is built) breaks down. In the experiment, reverse velocity inside the liquid film occurs and this also cannot be accommodated in the present model.

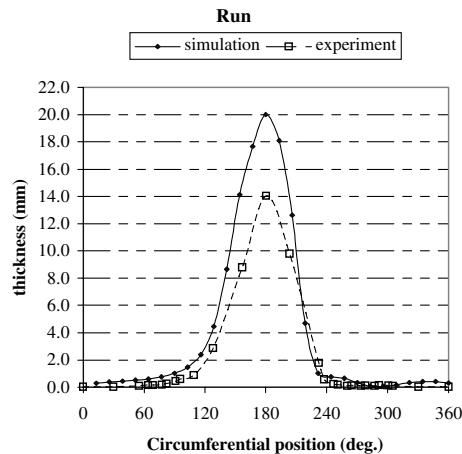


Fig. 17. Circumferential film thickness in run at gas split ratio 0.3.

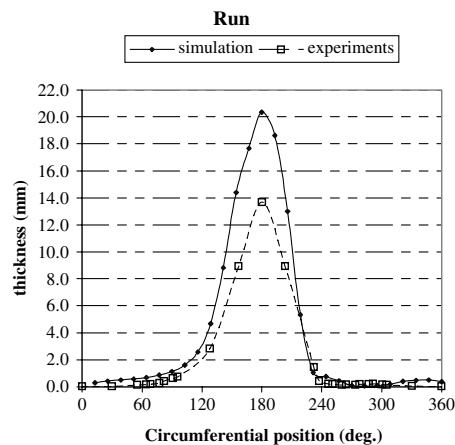


Fig. 18. Circumferential film thickness in run at gas split ratio 0.5.

Figs. 20–22 show the circumferential variation of the film thickness in the branch for different gas split ratios. Both in the calculation and experiment, two distinct peaks of film thickness can be discerned: one on the side of the branch facing the incoming flow and the other at the opposite inlet-branch corner.

At the side of the branch facing the incoming flow from the inlet of the T, as the gas split increases, the liquid split increases and therefore it is expected that at this location the film thickness increases: a peak of film thickness on the left hand side of Figs. 20–22 can be observed. However, the calculated location of this peak differs from the experiment. This is probably due to the neglect of the pressure variation within the film and which is bound to be significant in the vicinity of the junction; the method is unable to account for these variations and this is a weakness. At the corner inlet-branch, there is another peak in film thickness but in this case it is due to the gas phase re-circulation zone in this region: the liquid slows down and therefore the film

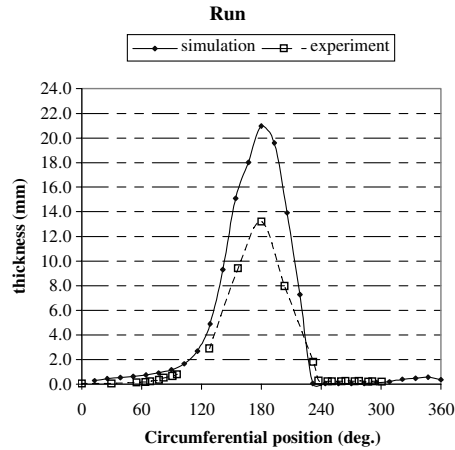


Fig. 19. Circumferential film thickness in run at gas split ratio 0.7.

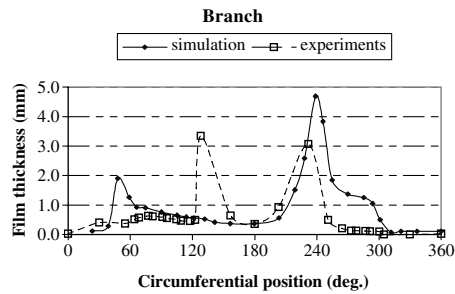


Fig. 20. Circumferential film thickness in branch at gas split ratio 0.5.

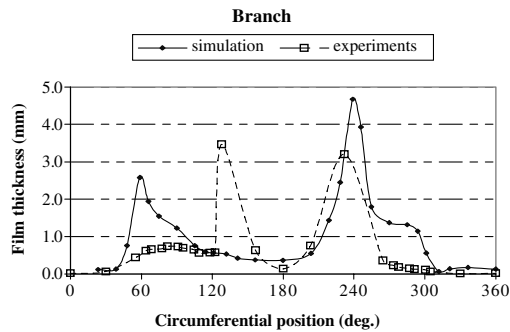


Fig. 21. Circumferential film thickness in branch at gas split ratio 0.3.

thickness increases. Fig. 23 shows the effect of increase in gas split on the film thickness at the branch of the T.

The most interesting from a practical point of view is the ability of the method to predict the degree of separation of the gas and liquid at the T. The numerically computed liquid split ratio as

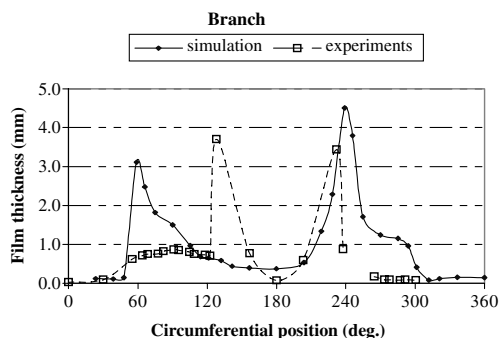


Fig. 22. Circumferential film thickness in branch at gas split 0.7.

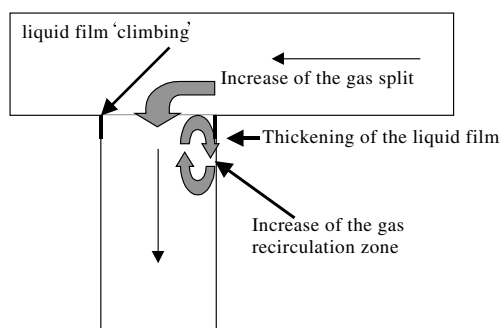


Fig. 23. Effect of the increase of the gas split on the film thickness at the branch.

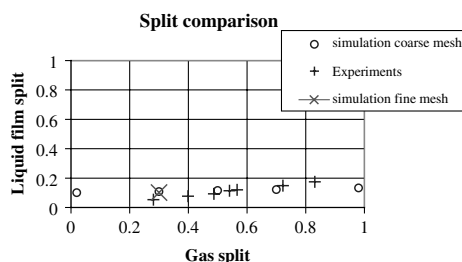


Fig. 24. Liquid split comparison.

a function of the gas split ratio for both the coarse and fine mesh computations are summarised in Fig. 24. In the figure, some of the measurements of Conte and Azzopardi [39] are also displayed. It should be pointed out that although simulations were carried out for a wide range of gas split ratios, the results can be relied upon only for ratios where the film remains fairly thin. In extremes of split ratios (greater than 0.8 and less than 0.2), the film thickness becomes quite large thereby invalidating the present model; hence a fair comparison of the method ought to focus on the range of split ratios between 0.3 and 0.7. In that range, it can be seen that the model and the experiment agree reasonably well in terms of the liquid split ratio. The simulation conducted with the

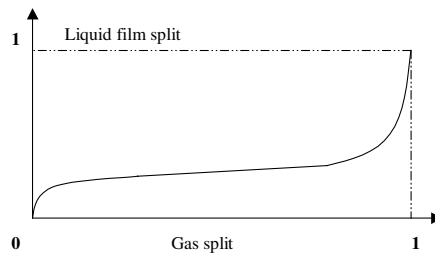


Fig. 25. Schematic representation of phase separation at T-junctions.

fine mesh gives very close results to the coarse mesh for one of the gas split ratios, which suggests that the results are independent of the mesh and are accurate enough for the problem. The low value of liquid split even with high gas split ratio suggests that the liquid film inertia is quite large and therefore the liquid always preferentially continues along the run. The model fails to predict the liquid film split at the extremes (when the gas split ratio approach zero or unity). In those cases, drastic decrease or increase of liquid film split as depicted in the sketch in Fig. 25 should result. However these are situations where the film thickness becomes very large thereby invalidating the assumptions built into the model. In such cases, alternative techniques, like interface tracking methods, must be used. In the mid-range of flow split ratios, however, where the thin film assumption holds, the present model appears to work remarkably well in predicting the split ratios.

4. Conclusions

A computational model for annular flow in pipes and pipe junctions has been presented. It comprises two sub-models, one for the dispersed droplet/gas flow in the core, and a separate model for the wall liquid film. The core model utilises a standard Lagrangian/Eulerian approach to solve the transport equations for the continuous gas and discrete droplet phases. The film model is based on simplifying assumptions, the main ones being: very thin film and boundary-layer type flow. The core and film models are coupled together through appropriate momentum and mass transfer terms at the interface. An extra force due to the form drag exerted by the gas flowing over the surface waves present on the film interface is included to account for the lift on the liquid film (which causes the observed wetting of the upper parts of horizontal tubes). Droplet entrainment from, and deposition to the film are also included in the model using standard empirical correlations.

Application of the model to horizontal fully developed pipe flow shows that the model is capable of predicting the film thickness and its velocity fairly accurately. The model is then applied to the prediction of the flow-split through a T-junction for which experimental data exist. Comparison with measurements reveal that the model is able to capture the broad features of the flow. In particular it is able to predict two distinct peaks in wall-film thickness that actually occur in the side branch of the T. However, it was unable to locate these peaks accurately. This is perhaps not surprising since the model does not account for the pressure distribution within the film; at the junction where the flow has to turn sharply, these effects are dominant and their

neglect could give rise to the observed discrepancies. Notwithstanding this shortcoming, it is shown that for a wide range of split ratios, the model is able to predict the degree of segregation of the phases occurring at the T with remarkable accuracy. In practical applications perhaps this is the most sought after quantity and here the model is able to provide a satisfactory answer.

Acknowledgements

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